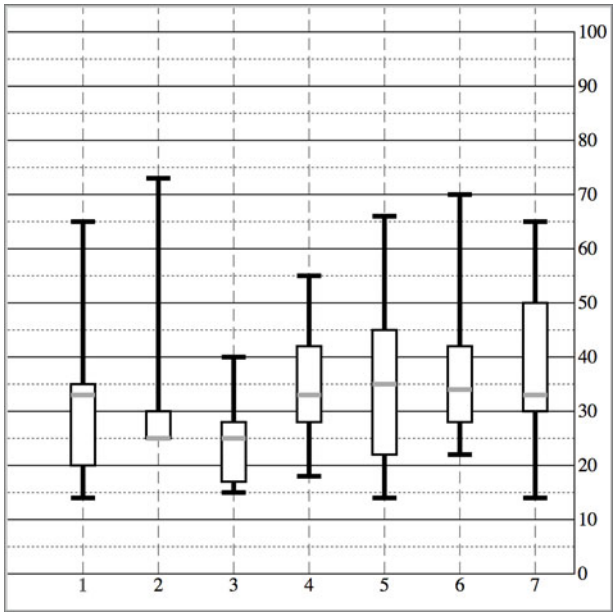


Figure 4.16 Our final boxplot chart. Each day now shows not only the median age of visitors but also the range of visiting ages, allowing for a more extensive examination of the demographics of site visitorship.

on a click of each boxplot, or make the width of the boxplot correspond to the number of visitors.

We looked at boxplots because a boxplot allows you to explore the creation of multipart objects while using lines and rectangles. But what's the value of a visualization like this that shows distribution? It encodes a graphical summary of the data, providing information about visitor age for the site on Wednesday, such as, "Most visitors were between the ages of 18 and 28. The oldest was 40. The youngest was 15. The median age was 25." It also allows you to quickly perform visual queries, checking to see if the median age of one day was within the majority of visitor ages of another day.

We'll stop exploring boxplots, and take a look at a different kind of complex graphical object: an interpolated line.

4.4 Line charts and interpolations

You create line charts by drawing connections between points. A line that connects points, and the shaded regions inside or outside the area constrained by the line, tell a story about the data. Although a line chart is technically a static data visualization, it's also a representation of change, typically over time.

We'll start with a new dataset in listing 4.7 that better represents change over time. Let's imagine we have a Twitter account and we've been tracking the number of tweets, favorites, and retweets to determine at what time we have the greatest response to our social media. Although we'll ultimately deal with this kind of data as JSON, we'll want to start with a comma-delimited file, because it's the most efficient for this kind of data.